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Mechatronics – ECEN 123/MECH 143/CSEN 123

Homework 3

Pre-Homework

Watch all the videos for this week. For each of the videos, write detail comments of what Maria is talking about. Maria expects to see the following headers so it is clear for which video you keep notes:

a) **Latches and Flipflops**
b) **Timers**
c) **PWM**
d) **Lab 3 Overview – Part 1**
e) **Lab 3 Overview – Part 2**

Maria expects very nice handwritten notes either on paper or tablet. Maria knows that some of you like to keep handwritten notes on the slides and this is totally ok to be submitted.

You will receive a 0 in the homework if (i) you do not have notes for all the videos and (ii) the notes are typed in and not handwritten. You will receive F in the class if you ask GenAI to read the transcript of the videos and generate the notes.

Scroll down for the homework problem.

First Notes I've written for this class btw

Latches & Flip-Flops

- Latches and flops are used to store data (bits)
- flops are clocked, latches aren't
- I know what pos/neg edges are
- More on this: In the video, she effectively uses the CLK signal as an enable for the latch. Latches are gate sensitive, not edge triggered, which is why you usually use an enable

Timers

- Use counters
- The functions are the same ones we wrote in prelab 3
- Must include library

PWM

- PWM = pulse width modulation
- Basically modifying duty cycle
- $\frac{\text{pulse width}}{\text{period}} \approx \frac{T_{on}}{T}$
- Simple and easy to use
- Use analogWrite (PWM) for motors
- Only 4 PWM pins at once
- Pin specs differ; refer to datasheet during implementation

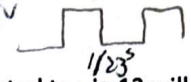
Lab 3 Overview

- 3 switches
  - on/off
  - two paths
  - pushbutton
- 90° diff means 90 deg phase
- We're gonna use a hex schmitt trigger
- Will use hall effect sensors
- Reed switch: magnetized switch
- Be gentle w/ motors

Problem 1

The code below is running on an Arduino board. Pin 3 is connected to a signal generator that is producing a 23Hz square wave at 5V amplitude.

5V



1/23s

Determine how many times the LED connected to pin 13 will flash on during each 10 second interval. You must explain how this is determined and show some math.

Note that there is a timer interrupt in this code, a pin interrupt in this code, and some polling code in loop().

```
#include <TimerOne.h>

bool inval;
bool lastval = false;

void setup()
{
  // Initialize the digital pin as an output.
  // Pin 13 has an LED connected on most Arduino boards

  pinMode(13, OUTPUT);
  pinMode(3, INPUT);
30Hz → Timer1.initialize(33333); // set a timer of length 33333 µseconds (or 0.033333 sec)
  Timer1.attachInterrupt( timerIsr );
  attachInterrupt(digitalPinToInterrupt(3), pulseIsr, FALLING);
}

void loop()
{
  inval = digitalRead(3);
  if ((inval != lastval) && (inval == true)) {
    digitalWrite( 13, digitalRead( 13 ) ^ 1 );
  }
  lastval = inval;
}

/// -------------------------
///  ISR Timer Routine
/// -------------------------
30Hz → void timerIsr()
{
    // Toggle LED
    digitalWrite( 13, digitalRead( 13 ) ^ 1 );
}

/// -------------------------
///  ISR Pin 3 Routine
/// -------------------------
23Hz → void pulseIsr()
{
    // Toggle LED
    digitalWrite( 13, digitalRead( 13 ) ^ 1 );
}
```

23Hz * 10 = 230 periods

falling edge

if just turned on, immediately turn back off.

Basically doubles Pin 3 routine flashes

10s = 300 periods

10s = 2300 periods

230 * 2 + 300 = ~~760 flashes~~

time on/off will be irregular, but b/c we XOR, we will always reach the expected flashes of

WRONG!

760 TOGGLES

$\frac{760}{2}$ = 380 flashes ✓

2(time interval)(Pin 3 f) + timer1 f (time interval)

= 2(10)(23) + 30(10) =

460 + 300 = 760